

MATH 431

Introduction to Probability

Midterm 2 Practice Book

University of Wisconsin–Madison

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Built from past Midterm 2 exams, Homework 5–9, and the course textbook

Heaviest signal: Fall 2023 (November 15, 2023) and Fall 2025 (November 19, 2025)

Name: _____

Goal before taking past exams cold:

- Get fast at identifying the right model.
- Get clean on MGFs, transformations, exponentials, and joint distributions.
- Practice the exact styles that keep reappearing on Midterm 2.

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How to Use This Book

This workbook is designed to be your main **practice-and-review bridge** before you take the actual past Midterm 2 exams under timed conditions. The problems start very gently, then ramp toward the structures that showed up repeatedly on the real exams.

Study Strategy

- Work the sections in order the first time. The sequencing is intentional: setup, then formulas, then transformations, then joint distributions, then mixed review.
- On the early problems, focus on **recognition and setup**. On the later problems, focus on writing full clean solutions without wasted steps.
- Leave answers in exact form whenever that is natural. Midterm 2 is not a calculator exam.
- When a problem says “approximate,” the important part is usually choosing and justifying the right approximation.
- After you finish Sections 4–10, take the actual past Midterm 2 exams cold.

How This Book Was Weighted

- **Very heavily weighted:** linear transformations of mean and variance, exponential distribution and memoryless property, MGFs, one-variable transformations, joint continuous distributions, marginals, independence, region probabilities, and joint expectations.
- **Also emphasized:** normal approximation, Poisson approximation, multinomial modeling, joint discrete tables, and simple geometry-based uniform models.
- **Included lightly:** confidence-interval/sample-size proportion questions, minimum/comparisons of exponentials, and non-monotone transformations such as $|X|$.
- **De-emphasized:** gamma-heavy material, hypergeometric-heavy material, hard decimals, and proof-heavy items that have not shown up as a main Midterm 2 pattern.

Main Textbook Sections Behind This Workbook

The core textbook support for this practice book is: Chapter 4 (especially Sections 4.1, 4.4, 4.5), Chapter 5 (Sections 5.1 and 5.2), Chapter 6 (Sections 6.1–6.3), and Chapter 8 (Sections 8.1–8.2 for expectation in the multivariate setting).

1 Distribution Recognition and Fast Setup

What You Need to Know

ROI: High

- **Binomial**: number of successes in n independent trials with the same success probability p .
- **Poisson**: counts of rare events; also the standard approximation to $\text{Bin}(n, p)$ when n is large, p is small, and $\lambda = np$.
- **Multinomial**: repeated independent trials with more than two mutually exclusive outcomes. The count vector has joint PMF

$$\mathbb{P}(X_1 = k_1, \dots, X_r = k_r) = \frac{n!}{k_1! \dots k_r!} p_1^{k_1} \dots p_r^{k_r}$$

whenever $k_1 + \dots + k_r = n$.

- **Exponential**: waiting time with constant rate λ ; mean $1/\lambda$.
- **Uniform on a region**: density is constant on the region, equal to $1/(\text{area of region})$ in two dimensions.
- **Normal**: appears directly, and also as an approximation after standardizing.

Exam Pattern / What This Usually Looks Like

These are usually quick points early in the exam: identify the right named distribution, write down the PMF or PDF, or decide whether a count vector is binomial, Poisson, or multinomial before doing any real computation.

Warm-Up Reminder

If each trial has *three* mutually exclusive outcomes with fixed probabilities and the trials are independent, then the count vector is **multinomial**, not binomial.

Where to Review More

Textbook Sections 3.1, 4.4, 4.5, 6.1, 6.2.

Level 1: Warm-Up

Problem 1.1**[Warm-up]**

A quiz app is shown to 25 users. Each user independently answers “yes” with probability 0.4. Let X be the number of “yes” answers.

1. What distribution does X have?
2. Write its PMF.
3. State $\mathbb{E}[X]$ and $\text{Var}(X)$.

Problem 1.2**[Warm-up]**

Each manufactured chip is independently classified as *acceptable*, *rework*, or *scrap* with probabilities 0.92, 0.06, and 0.02, respectively. Let (X_1, X_2, X_3) be the counts of these three outcomes among 50 chips.

1. What named distribution models (X_1, X_2, X_3) ?
2. Write the joint PMF.

Level 2: Standard**Problem 1.3****[Standard]**

A sensor lifetime is modeled by an exponential distribution with expected lifetime 8 months.

1. Find the rate parameter λ .
2. Write the CDF.
3. Compute $\mathbb{P}(X > 10)$.

Problem 1.4**[Standard]**

A point is chosen uniformly at random from the rectangle $[0, 3] \times [0, 2]$.

1. Write the joint PDF of (X, Y) .
2. Compute $\mathbb{P}(X < 1, Y > 1)$.

Level 3: Exam-Style**Problem 1.5****[Exam-style]**

A streaming platform classifies each of 400 visits as desktop, mobile, or tablet with probabilities 0.55, 0.35, and 0.10, independently across visits.

1. What distribution models the count vector (D, M, T) ?
2. Write the exact probability that there are exactly 210 desktop visits, 145 mobile visits, and 45 tablet visits.

2 Expectation, Variance, and Linear Transformations

What You Need to Know

ROI: Extremely High

- For constants a, b ,

$$\mathbb{E}[aX + b] = a \mathbb{E}[X] + b, \quad \text{Var}(aX + b) = a^2 \text{Var}(X).$$

- If you want $Y = aX + b$ to have target mean m and target variance s^2 , solve

$$a \mathbb{E}[X] + b = m, \quad a^2 \text{Var}(X) = s^2.$$

- Quick reminders:

$$\mathbb{E}[\text{Bin}(n, p)] = np, \quad \text{Var}(\text{Bin}(n, p)) = np(1 - p),$$

$$\mathbb{E}[\text{Poisson}(\lambda)] = \lambda, \quad \text{Var}(\text{Poisson}(\lambda)) = \lambda,$$

$$\mathbb{E}[\text{Exp}(\lambda)] = 1/\lambda, \quad \text{Var}(\text{Exp}(\lambda)) = 1/\lambda^2.$$

Exam Pattern / What This Usually Looks Like

This is one of the most reliable short-problem types on Midterm 2. It is usually fast algebra, but it is easy to lose points by forgetting that the variance uses a^2 , not a .

Warm-Up Reminder

If $\mathbb{E}[X] = 2$ and $\text{Var}(X) = 3$, then $\mathbb{E}[3X - 1] = 5$ and $\text{Var}(3X - 1) = 27$.

Where to Review More

Textbook Sections 3.3, 3.4, and 8.1.

Level 1: Warm-Up

Problem 2.1

[Warm-up]

Suppose $\mathbb{E}[X] = 3$ and $\text{Var}(X) = 5$. Find

$$\mathbb{E}[2X - 7] \quad \text{and} \quad \text{Var}(2X - 7).$$

Problem 2.2**[Warm-up]**

Let $X \sim \text{Bin}(60, 1/5)$.

1. Find $\mathbb{E}[X]$ and $\text{Var}(X)$.
2. Find $\mathbb{E}[4 - X]$ and $\text{Var}(4 - X)$.

Level 2: Standard**Problem 2.3****[Standard]**

Suppose $\mathbb{E}[X] = -2$ and $\text{Var}(X) = 9$. Find numbers a, b so that $aX + b$ has mean 0 and variance 1.

Problem 2.4**[Standard]**

Suppose $\mathbb{E}[X] = 5$ and $\text{Var}(X) = 4$. Find numbers a, b so that $aX + b$ has mean 1 and variance 36.

Problem 2.5**[Standard]**

Let $S = 300 + W$ where $W \sim \text{Bin}(100, 1/2)$.

1. Find $\mathbb{E}[S]$.
2. Find $\text{Var}(S)$.
3. State the standard deviation of S .

Level 3: Exam-Style

Problem 2.6**[Exam-style]**

Let X be a random variable with mean -6 and variance 25 . Find positive numbers a, b so that the random variable

$$Y = aX + b$$

has mean 0 and variance 4 .

3 Exponential Distributions and the Memoryless Property

What You Need to Know

ROI: Extremely High

- If $X \sim \text{Exp}(\lambda)$, then

$$f_X(x) = \lambda e^{-\lambda x} \quad (x > 0), \quad F_X(x) = 1 - e^{-\lambda x} \quad (x \geq 0).$$

- Tail probability:

$$\mathbb{P}(X > t) = e^{-\lambda t}.$$

- Mean and variance:

$$\mathbb{E}[X] = 1/\lambda, \quad \text{Var}(X) = 1/\lambda^2.$$

- Memoryless property:

$$\mathbb{P}(X > s + t \mid X > s) = \mathbb{P}(X > t).$$

- If independent exponentials have rates $\lambda_1, \dots, \lambda_n$, then the minimum has exponential rate $\lambda_1 + \dots + \lambda_n$.

Exam Pattern / What This Usually Looks Like

The exam version is usually one of these: recover the rate from the mean, compute a direct tail probability, use memorylessness for a conditional probability, or compare two independent exponentials.

Warm-Up Reminder

If the expected waiting time is 12, then the rate is $\lambda = 1/12$, not 12.

Where to Review More

Textbook Section 4.5.

Level 1: Warm-Up

Problem 3.1**[Warm-up]**

Let $X \sim \text{Exp}(1/5)$. Compute:

$$\mathbb{P}(X < 5) \quad \text{and} \quad \mathbb{P}(X > 8).$$

Problem 3.2**[Warm-up]**

A battery life is modeled by an exponential distribution with expected life 18 months.

1. Find the rate parameter.
2. Compute $\mathbb{P}(X < 18)$.
3. Compute $\mathbb{P}(X > 24)$.

Level 2: Standard**Problem 3.3****[Standard]**

A phone call length is modeled by an exponential distribution with mean 12 minutes. Find

$$\mathbb{P}(X > 20 \mid X > 8).$$

Problem 3.4**[Standard]**

Let $X \sim \text{Exp}(\lambda)$. Show that

$$\mathbb{P}(X > s + t \mid X > s)$$

does not depend on s , and simplify it fully.

Problem 3.5**[Standard]**

Let $X \sim \text{Exp}(2)$ and $Y \sim \text{Exp}(3)$ be independent. Find $\mathbb{P}(X < Y)$.

Level 3: Exam-Style**Problem 3.6****[Exam-style]**

Let $\lambda > 0$. Suppose $X \sim \text{Exp}(\lambda)$ and $Y \sim \text{Exp}(2\lambda)$ are independent.

1. Compute $\mathbb{P}(X > 3Y)$.
2. Let $V = \min(X, Y)$. Find the density $f_V(v)$.

4 Normal Approximation, Poisson Approximation, and Multinomial Modeling

What You Need to Know

ROI: Extremely High

- If $X \sim \text{Bin}(n, p)$ and $np(1 - p) > 10$, then

$$\frac{X - np}{\sqrt{np(1 - p)}}$$

is well approximated by a standard normal random variable.

- Continuity correction often improves normal approximations for integer-valued X .
- If $X \sim \text{Bin}(n, p)$ with n large and p small, then a Poisson approximation with $\lambda = np$ is good when np^2 is small.
- A count vector over several mutually exclusive categories is multinomial, not “several independent binomials.”
- Conservative proportion interval rule:

$$\hat{p} \pm \frac{1.96}{2\sqrt{n}}$$

comes from the bound $p(1 - p) \leq 1/4$.

Exam Pattern / What This Usually Looks Like

You are usually expected to do two things: choose the right approximation or model, and justify why it is appropriate. The justification is often as important as the final expression.

Warm-Up Reminder

If $X \sim \text{Bin}(400, 0.1)$, then $np(1 - p) = 36$, so normal approximation is sensible. If $X \sim \text{Bin}(10000, 0.001)$, then $np^2 = 0.01$, so Poisson approximation is excellent.

Where to Review More

Textbook Sections 4.1, 4.3, 4.4, and 6.1.

Level 1: Warm-Up

Problem 4.1**[Warm-up]**

Let $X \sim \text{Bin}(240, 0.3)$.

1. State the mean and variance.
2. Write a normal approximation for $\mathbb{P}(X \leq 80)$ in terms of Φ .
3. Also write the continuity-corrected version.

Problem 4.2**[Warm-up]**

Let $X \sim \text{Bin}(5000, 0.0008)$. Decide whether normal or Poisson approximation is more natural for approximating $\mathbb{P}(X = 2)$, and then write the approximation.

Level 2: Standard**Problem 4.3****[Standard]**

A promotion places a silver coupon in each box with probability 0.08, independently across boxes. A customer buys 300 boxes. If X is the number of silver coupons found, approximate

$$\mathbb{P}(X \geq 30).$$

Leave the answer in terms of Φ .

Problem 4.4**[Standard]**

A rare defect occurs independently with probability 0.002 on each of 2500 items. Approximate the probability of at least one defect.

Problem 4.5**[Standard]**

A class has 180 students, and each student is left-handed independently with probability 0.12. Approximate the probability that more than 25 students are left-handed.

Problem 4.6**[Standard]**

Derive a sufficient sample size n for a 95% confidence interval for a population proportion to have **total width** at most w , using the conservative bound $p(1 - p) \leq 1/4$.

Problem 4.7**[Standard]**

A survey of size $n = 900$ finds sample proportion $\hat{p} = 0.18$. Write the conservative approximate 95% interval.

Level 3: Exam-Style**Problem 4.8****[Exam-style]**

Each package contains a bronze card with probability 0.12, a gold card with probability 0.005, or no card otherwise. Assume packages are independent. A store sells 400 packages.

1. Approximate the probability of at least 60 bronze cards.
2. Approximate the probability of exactly 2 gold cards.
3. Write the exact probability of getting exactly 55 bronze cards and exactly 1 gold card.

5 Moment Generating Functions

What You Need to Know

ROI: Extremely High

- The moment generating function is

$$M_X(t) = \mathbb{E}[e^{tX}].$$

- If M_X is finite around 0, then

$$M'_X(0) = \mathbb{E}[X], \quad M''_X(0) = \mathbb{E}[X^2], \quad \text{Var}(X) = M''_X(0) - (M'_X(0))^2.$$

- If X is discrete with finitely many values, then

$$M_X(t) = \sum_x e^{tx} \mathbb{P}(X = x),$$

so you can often **read off** the support and probabilities.

- Useful recognition patterns:

$$M_{\text{Bin}(n,p)}(t) = ((1-p) + pe^t)^n,$$

$$M_{\text{Poisson}(\lambda)}(t) = e^{\lambda(e^t-1)},$$

$$M_{\text{Exp}(\lambda)}(t) = \frac{\lambda}{\lambda - t} \quad (t < \lambda).$$

Exam Pattern / What This Usually Looks Like

The exam loves two MGF skills: recover a PMF from a finite sum of exponential terms, and use derivatives at 0 to compute moments or variance.

Warm-Up Reminder

$M'''_X(0) = \mathbb{E}[X^3]$. If the MGF is already written as a weighted sum of terms like e^{tx} , it is often faster to read off the distribution first and then compute moments from the PMF.

Where to Review More

Textbook Section 5.1 and Section 8.3.

Level 1: Warm-Up

Problem 5.1**[Warm-up]**

Suppose

$$M_X(t) = \frac{1}{4}e^{-t} + \frac{1}{2}e^t + \frac{1}{4}e^{3t}.$$

Write the PMF of X .**Problem 5.2****[Warm-up]**For the random variable in the previous problem, find $\mathbb{P}(X > 0)$ and $\mathbb{E}[X]$.**Level 2: Standard****Problem 5.3****[Standard]**

Suppose

$$M_X(t) = \frac{1}{8} + \frac{3}{8}e^t + \frac{3}{8}e^{2t} + \frac{1}{8}e^{3t}.$$

1. Find the PMF of X .
2. Identify the distribution, including parameters.

Problem 5.4**[Standard]**

Let

$$M_X(t) = \left(\frac{1+e^t}{2}\right)^4.$$

Find $\mathbb{E}[X]$, $\text{Var}(X)$, and $\mathbb{P}(X \geq 3)$.

Problem 5.5**[Standard]**

Let

$$M_X(t) = \frac{1}{\sqrt{1-2t}} \quad \text{for } |t| < 1/2.$$

Find $\mathbb{E}[X]$ and $\text{Var}(X)$.**Problem 5.6****[Standard]**

Let

$$M_X(t) = e^{2(e^t-1)}.$$

Identify the distribution of X , then compute $\mathbb{P}(X = 0)$, $\mathbb{E}[X]$, and $\text{Var}(X)$.**Level 3: Exam-Style****Problem 5.7****[Exam-style]**

Suppose

$$M_X(t) = \frac{1}{10}e^{-2t} + \frac{3}{10} + \frac{2}{5}e^t + \frac{1}{5}e^{4t}.$$

Find:

1. the PMF of X ,
2. $\mathbb{P}(X \text{ is even})$,
3. $M_X'''(0)$ without doing unnecessary symbolic differentiation,
4. $\text{Var}(X)$.

6 Transformations of One Continuous Random Variable

What You Need to Know

ROI: Extremely High

- **Support first.** Before any algebra, determine the possible values of $Y = g(X)$.
- **CDF method:**

$$F_Y(y) = \mathbb{P}(g(X) \leq y).$$

This is often the safest method, especially for non-monotone functions.

- If g is one-to-one and differentiable, then

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|.$$

- If g is not one-to-one, add contributions from all relevant branches.
- Standard trouble spots:
 - forgetting to split the support for $|X|$ or X^2 ,
 - differentiating before finding the correct support,
 - losing absolute values in the derivative.

Exam Pattern / What This Usually Looks Like

This is one of the biggest Midterm 2 themes. Real exam versions often use a uniform random variable and ask for the density of X^2 , X^3 , $|X|$, or a geometry-based quantity such as area.

Warm-Up Reminder

If $X \sim \text{Unif}[0, b]$ and $Y = X^2$, then on $0 \leq y \leq b^2$,

$$F_Y(y) = \mathbb{P}(X \leq \sqrt{y}) = \frac{\sqrt{y}}{b}.$$

Where to Review More

Textbook Section 5.2 and Homework 5 / Homework 8.

Level 1: Warm-Up

Problem 6.1**[Warm-up]**

Let $X \sim \text{Unif}[0, 4]$ and let $Y = X^2$.

1. Find the support of Y .
2. Find $F_Y(y)$ for all real y .
3. Find a density $f_Y(y)$.

Problem 6.2**[Warm-up]**

Let $X \sim \text{Unif}[-2, 4]$ and let $Y = X^3$. Find a density $f_Y(y)$.

Level 2: Standard**Problem 6.3****[Standard]**

Let $X \sim \text{Unif}[-2, 4]$ and let $Y = |X|$. Find a density $f_Y(y)$.

Problem 6.4**[Standard]**

Suppose X has density

$$f_X(x) = 2x, \quad 0 \leq x \leq 1.$$

Find the density of $Y = 1 - X$.

Problem 6.5**[Standard]**

Suppose X has density

$$f_X(x) = 3x^2, \quad 0 \leq x \leq 1.$$

Find the density of $Y = \sqrt{X}$.

Problem 6.6**[Standard]**

A point is thrown uniformly at random onto a disk of radius 6. Let R be its distance from the center and let

$$A = \pi R^2$$

be the area of the centered disk through the point.

1. Find the support of A .
2. Find $F_A(a)$.
3. Find a density $f_A(a)$.

Level 3: Exam-Style**Problem 6.7****[Exam-style]**

Let $X \sim \text{Exp}(1)$ and define

$$Y = (X - 2)^2.$$

1. Find $F_Y(y)$ for all $y \geq 0$.
2. Find a density $f_Y(y)$.

7 Joint Discrete Distributions

What You Need to Know

ROI: High

- Marginals come from row sums and column sums.
- Independence means

$$p_{X,Y}(x, y) = p_X(x)p_Y(y)$$

for all pairs (x, y) in the support.

- To show *not* independent, one counterexample is enough.
- Probabilities of events come from summing the relevant entries.
- Expectations come from

$$\mathbb{E}[g(X, Y)] = \sum_x \sum_y g(x, y) p_{X,Y}(x, y).$$

Exam Pattern / What This Usually Looks Like

The common exam table problem asks for marginals, then independence, then a short probability or variance computation. These are usually fast points if your table-reading is clean.

Warm-Up Reminder

If a single table entry is 0 while the corresponding product of marginals is positive, then independence fails immediately.

Where to Review More

Textbook Sections 6.1 and 6.3, plus Homework 9.

Level 1: Warm-Up

Problem 7.1

[Warm-up]

The joint PMF of (X, Y) is given by the table

$X \backslash Y$	0	1	2
0	$\frac{1}{10}$	$\frac{2}{10}$	$\frac{1}{10}$
1	$\frac{2}{10}$	$\frac{1}{10}$	$\frac{3}{10}$

Find the marginal PMFs of X and Y .

Problem 7.2**[Warm-up]**

For the joint table in the previous problem, determine whether X and Y are independent. Give a rigorous explanation.

Level 2: Standard**Problem 7.3****[Standard]**

For the same joint table, compute $\mathbb{P}(Y > X)$ and $\mathbb{E}[Y]$.

Problem 7.4**[Standard]**

The joint PMF of (X, Y) is given by

$X \backslash Y$	0	1
1	$\frac{1}{8}$	$\frac{1}{8}$
2	$\frac{1}{8}$	$\frac{1}{4}$
3	$\frac{1}{8}$	$\frac{1}{4}$

Compute:

1. $\mathbb{E}[XY]$,
2. $\mathbb{P}(X = Y + 1)$.

Level 3: Exam-Style

Problem 7.5**[Exam-style]**

An urn experiment is repeated n times independently. Each draw is red with probability $1/2$, blue with probability $1/3$, and green with probability $1/6$. Let (R, B, G) be the counts.

1. Write the joint PMF of (R, B, G) .
2. When $n = 6$, compute $\mathbb{P}(R = 4, B = 2, G = 0)$.
3. State the marginal distribution of R .

8 Joint Continuous Distributions: Marginals, Independence, and Region Probabilities

What You Need to Know

ROI: Extremely High

- A valid joint PDF must satisfy $f(x, y) \geq 0$ and

$$\iint_{\mathbb{R}^2} f(x, y) dA = 1.$$

- Marginals:

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy, \quad f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx.$$

- Independence means

$$f(x, y) = f_X(x)f_Y(y)$$

on the support.

- If (X, Y) is uniform on a region D , then

$$f(x, y) = \frac{1}{\text{area}(D)}$$

on D and 0 outside.

- For region probabilities, draw the support and the event region before integrating.

Exam Pattern / What This Usually Looks Like

This is the other huge Midterm 2 theme. Expect marginals, independence, a region probability, and often a support where the geometry matters just as much as the calculus.

Warm-Up Reminder

If the support is triangular, independence is usually impossible because products of one-variable densities naturally live on rectangles, not triangles.

Where to Review More

Textbook Sections 6.2 and 6.3.

Level 1: Warm-Up

Problem 8.1**[Warm-up]**

Find the constant c so that

$$f(x, y) = c(x + y), \quad 0 \leq x \leq 1, \ 0 \leq y \leq 1,$$

is a joint PDF.

Problem 8.2**[Warm-up]**

For the joint PDF in the previous problem, find $f_X(x)$ and $f_Y(y)$.

Level 2: Standard**Problem 8.3****[Standard]**

For the joint PDF

$$f(x, y) = c(x + y), \quad 0 \leq x \leq 1, \ 0 \leq y \leq 1,$$

with the correct normalizing constant c , determine whether X and Y are independent.

Problem 8.4**[Standard]**

For the same joint PDF, compute $\mathbb{P}(X < Y)$.

Problem 8.5**[Standard]**

Suppose (X, Y) is uniformly distributed on the triangle with vertices $(0, 0)$, $(2, 0)$, and $(2, 1)$.

1. Write the joint PDF.
2. Find $f_X(x)$ and $f_Y(y)$.

Problem 8.6**[Standard]**

For the triangle in the previous problem, compute $\mathbb{P}(X + Y < 2)$.

Problem 8.7**[Standard]**

Let

$$f(x, y) = 6e^{-2x-3y}, \quad x > 0, y > 0,$$

and 0 otherwise.

1. Check whether X and Y are independent.
2. Identify the marginal densities.

Level 3: Exam-Style

Problem 8.8**[Exam-style]**

Let

$$f(x, y) = 3x, \quad 0 < y < x < 1,$$

and 0 otherwise.

1. Find $f_X(x)$ and $f_Y(y)$.
2. Determine whether X and Y are independent.
3. Compute $\mathbb{P}(Y \leq 1/2, X > 3/4)$.

9 Joint Expectations, Double Integrals, and Uniform-on-Region Geometry

What You Need to Know

ROI: Extremely High

- In the joint continuous setting,

$$\mathbb{E}[g(X, Y)] = \iint g(x, y) f(x, y) dA$$

whenever the integral is well-defined.

- The same framework gives probabilities:

$$\mathbb{P}((X, Y) \in A) = \iint_A f(x, y) dA.$$

- If a question asks only for an *integral expression*, the setup is the answer.
- On uniform regions, geometry often gives the cleanest path to support and density before you integrate.

Exam Pattern / What This Usually Looks Like

Expect one of the following: write down a correct double integral over a region, compute $\mathbb{E}[X]$, $\mathbb{E}[Y]$, or $\mathbb{E}[XY]$, or use a uniform-on-region model where the geometry is simple but the setup must be exact.

Warm-Up Reminder

If the problem says “Do not compute; only set up the integral,” do exactly that. Good limits and a correct integrand are full credit material on this type of question.

Where to Review More

Textbook Sections 6.2, 8.1, and 8.2.

Level 1: Warm-Up

Problem 9.1**[Warm-up]**

Suppose (X, Y) has joint PDF $A(x, y)$ on the rectangle $0 \leq x \leq 4$, $0 \leq y \leq 3$, and 0 outside. Write integral expressions for:

1. the condition that A is a valid joint PDF,
2. $\mathbb{E}[X^2Y]$,
3. $\mathbb{P}(X + Y < 2)$.

Level 2: Standard**Problem 9.2****[Standard]**

Suppose (X, Y) is uniformly distributed on the triangle with vertices $(0, 0)$, $(1, 0)$, and $(1, 2)$.

1. Compute $\mathbb{E}[X]$.
2. Compute $\mathbb{E}[Y]$.

Problem 9.3**[Standard]**

For the same triangle, compute $\mathbb{E}[XY]$.

Problem 9.4**[Standard]**

Let

$$f(x, y) = xe^{-x(1+y)}, \quad x > 0, y > 0,$$

and 0 otherwise. Compute $\mathbb{E}[XY]$.**Problem 9.5****[Standard]**A point is chosen uniformly on the triangle with vertices $(0, 0)$, $(2, 0)$, and $(0, 2)$.

1. Compute $\mathbb{P}(X + Y < 1)$.
2. Compute $\mathbb{E}[X + Y]$.

Level 3: Exam-Style**Problem 9.6****[Exam-style]**

Let

$$f(x, y) = 1, \quad 0 < x < 1, x < y < x + 1,$$

and 0 otherwise.

1. Find $f_X(x)$ and $f_Y(y)$.
2. Determine whether X and Y are independent.
3. Write an integral expression for $\mathbb{P}(X > 1/3, Y < 5/4)$.

10 Mixed Exam-Style Review

How to Use This Section

ROI: Extremely High Treat these as dress rehearsals. Try to decide the model quickly, write a clean setup, and avoid overcomputing. These are deliberately built to resemble actual Midterm 2 pacing and structure without copying the old exams directly.

Mixed Set

Problem 10.1

[Exam-style]

Each of 600 snack bags contains a silver sticker with probability 0.09, a gold sticker with probability 0.001, or no sticker otherwise. Assume bags are independent.

1. Approximate the probability of at least 65 silver stickers.
2. Approximate the probability of at least one gold sticker.
3. Write the exact probability of exactly 58 silver stickers and exactly 2 gold stickers.

Problem 10.2

[Exam-style]

A random variable X has moment generating function

$$M_X(t) = \frac{1}{16}(1 + e^t)^4.$$

1. Derive the PMF of X .
2. Find $\mathbb{E}[X]$ and $\text{Var}(X)$.
3. Let $Y = 2X - 5$. Find $\mathbb{E}[Y]$, $\text{Var}(Y)$, and $\mathbb{P}(Y > 1)$.

Problem 10.3**[Exam-style]**

A point is chosen uniformly at random on a disk of radius 4. Let T be the area of the centered disk through the chosen point.

1. Find the support of T .
2. Find the cumulative distribution function $F_T(t)$.
3. Find a density $f_T(t)$.
4. Compute $\mathbb{P}(5\pi < T < 10\pi)$.

Problem 10.4**[Exam-style]**

Suppose (X, Y) is uniformly distributed on the triangle with vertices $(0, 0)$, $(2, 0)$, and $(2, 2)$.

1. Write the joint PDF.
2. Find $f_X(x)$ and $f_Y(y)$.
3. Determine whether X and Y are independent.
4. Compute $\mathbb{P}(X < 1, Y > 1/2)$.
5. Compute $\mathbb{E}[XY]$.

Problem 10.5**[Exam-style]**

Let $\lambda > 0$. Suppose $X \sim \text{Exp}(\lambda)$ and $Y \sim \text{Exp}(3\lambda)$ are independent.

1. Compute $\mathbb{P}(X > 2Y)$.
2. Let $V = \min(X, Y)$. Find the density of V .
3. Compute $\mathbb{P}(V > t)$ for $t > 0$.

Final Notes Before Past Exams

What to Be Fluent On

- Solve $aX + b$ mean/variance questions almost automatically.
- Move comfortably between PMF/PDF, CDF, MGF, and support.
- Know when to use normal approximation, Poisson approximation, and multinomial modeling.
- Be able to set up region integrals from a picture.
- Be able to get marginals and test independence from both joint tables and joint densities.
- For exponentials, do not miss the mean-rate conversion or the memoryless step.

Suggested Next Step

After you can do most of this workbook without hesitation, take the real past Midterm 2 exams under realistic timing. Mark every miss by category: setup mistake, algebra mistake, approximation choice, support error, or integration-region error. That is the fastest way to clean up the last weak spots.